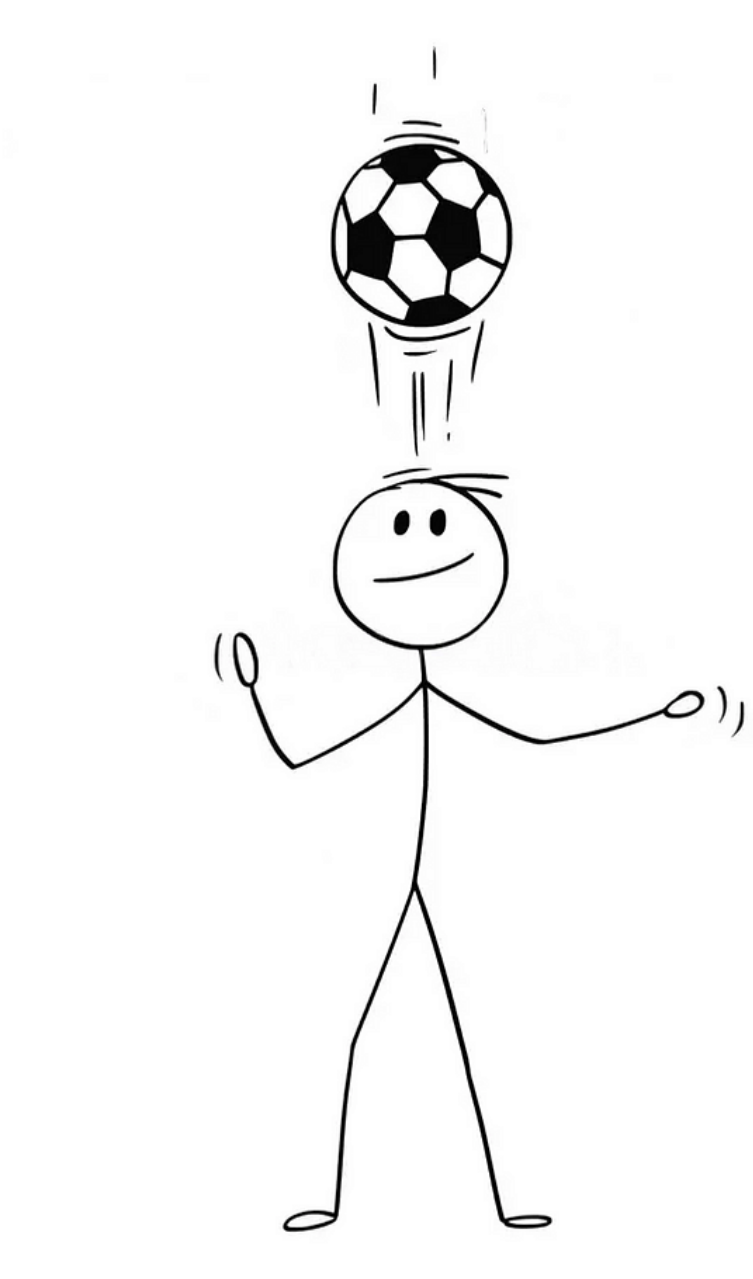
< Go to the original



I Spent a Week Studying HTTP Headers—Here's What I Learnt

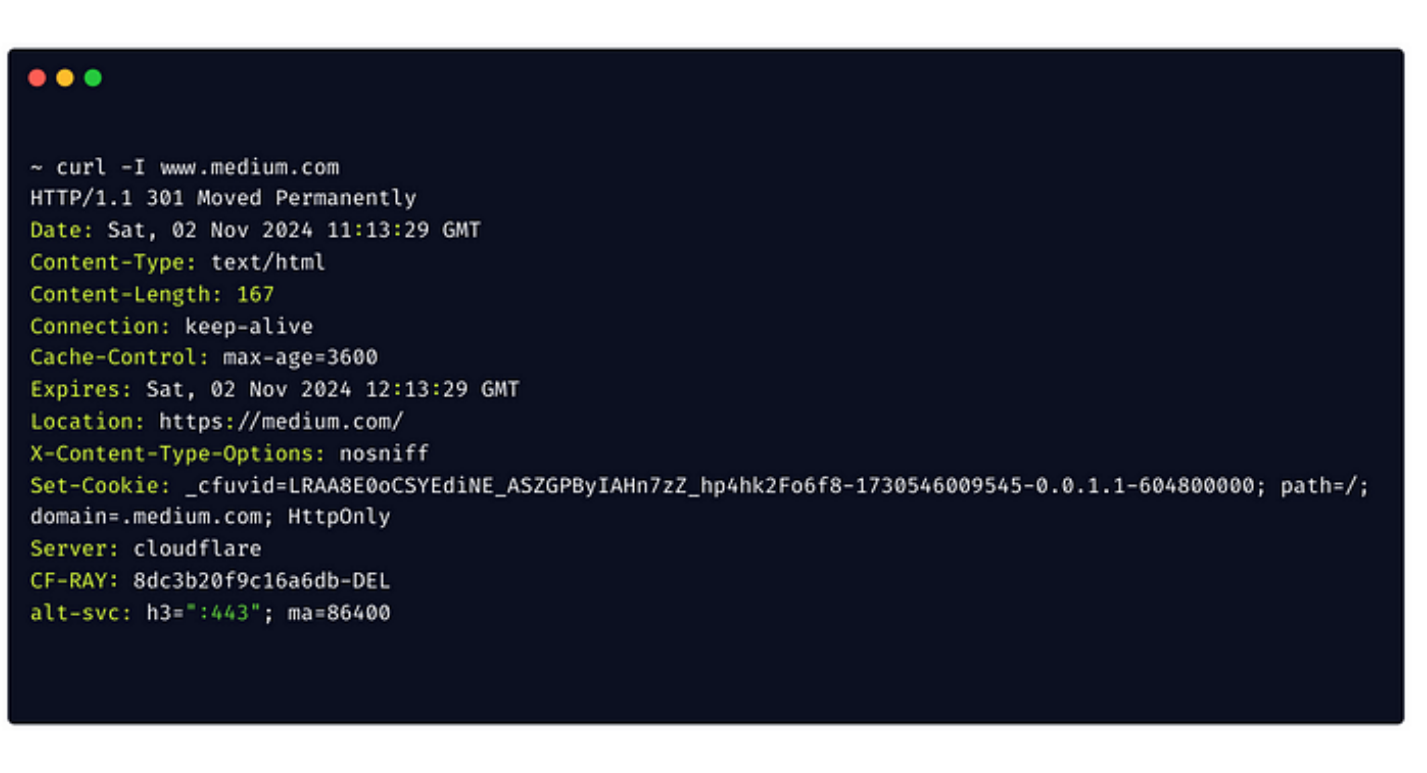
They're in Every Request We Make, Yet Most Developers Overlook Their Power

Rishabh Agarwal
Follow

Level Up Coding a11y-light ~7 min read · November 5, 2024 (Updated: November 5, 2024) ·
Free: No

The modern web is extensively built upon HTTP requests and responses. Each HTTP transfer is accompanied by several HTTP headers containing information about the client, the server, the requested content, the payload, and more.

An HTTP headers consists of a case-insensitive name followed by a colon, and then by its value. Whitespaces before the header value is ignored! And, a single HTTP request can have more than one HTTP headers as well.



```
~ curl -I www.medium.com
HTTP/1.1 301 Moved Permanently
Date: Sat, 02 Nov 2024 11:13:29 GMT
Content-Type: text/html
Content-Length: 167
Connection: keep-alive
Cache-Control: max-age=3600
Expires: Sat, 02 Nov 2024 12:13:29 GMT
Location: https://medium.com/
X-Content-Type-Options: nosniff
Set-Cookie: _cfuvid=LRAA8E0oCSYEdiNE_ASZGPByIAHn7zZ_hp4hk2Fo6f8-1730546009545-0.0.1.1-604800000; path=/;
domain=.medium.com; HttpOnly
Server: cloudflare
CF-RAY: 8dc3b20f9c16a6db-DEL
alt-svc: h3=":443"; ma=86400
```

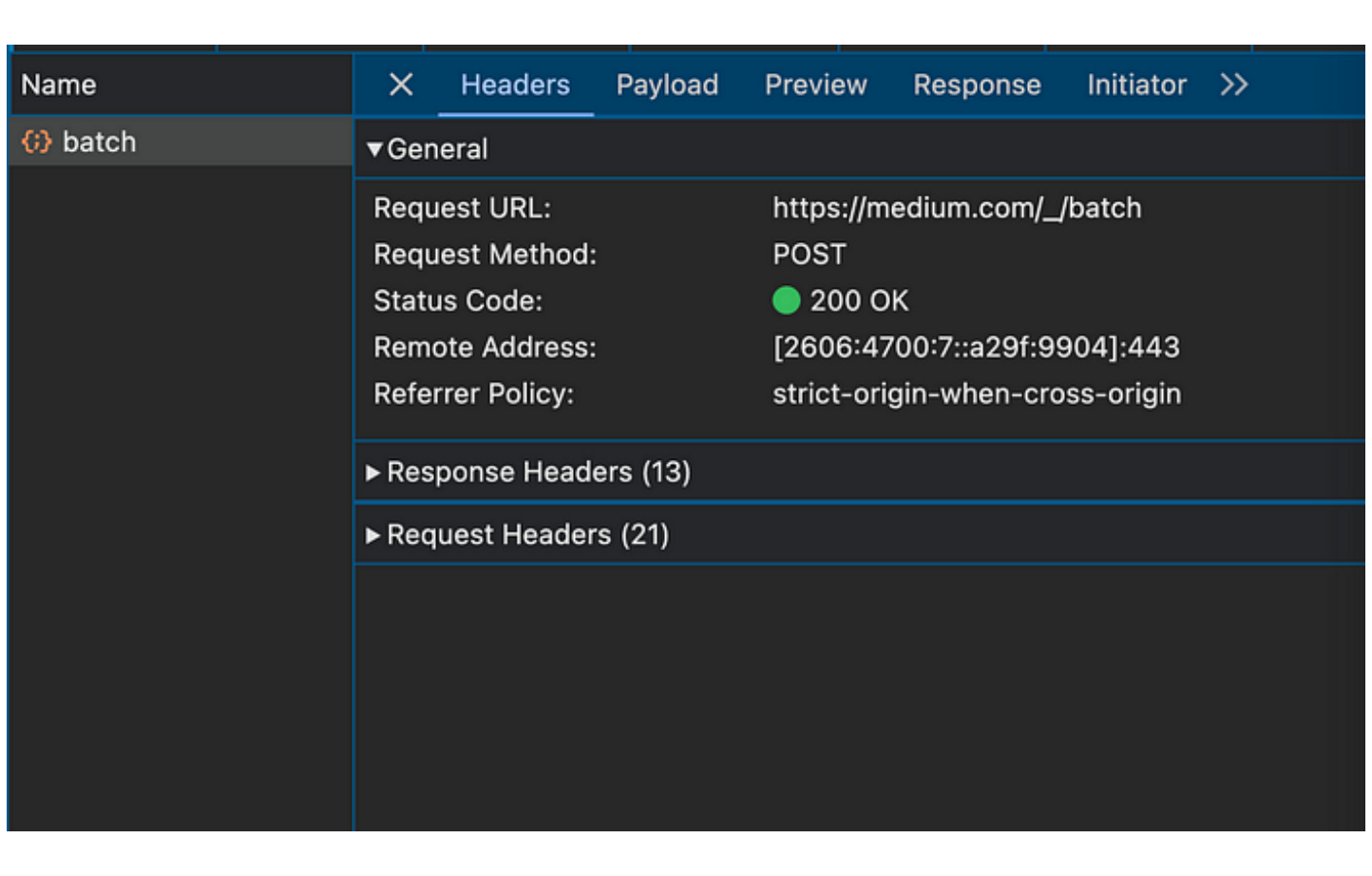
While headers might seem invisible to end-users who primarily interact with the payload (the actual content), they play a crucial role in web applications. For us, the developers, understanding these headers is essential for several reasons:

- *Modern browsers rely heavily on these headers to determine how to process and display server responses. For instance, cache-control headers dictate whether to fetch fresh content or use cached data, while content-type headers tell browsers how to interpret the received data.*
- *Servers make critical decisions based on request headers. Authentication tokens, client capabilities, and preferred content formats — all transmitted through headers — can significantly influence how a server handles each request.*
- *Headers consume bandwidth. Each additional header increases the size of HTTP messages, impacting network performance. Unnecessary headers can accumulate to create noticeable delays, especially on slower networks or when dealing with high-volume traffic.*

In the remainder of this article, we will take a deep-dive into the topic of HTTP headers and understand how different kinds of headers are used in HTTP communications!

How to See HTTP Headers?

There are multiple ways to inspect the headers associated with an HTTP request, but using a browser's developer tools is by far the most preferred method. When a network request is made on a webpage, it appears in the Network tab of the Developer Tools. From there, you can easily access the request and response headers, along with other general information about any network call made from the browser page.



Request URL: https://medium.com/_/batch
Request Method: POST
Status Code: ● 200 OK
Remote Address: [2606:4700:7::a29f:9904]:443
Referrer Policy: strict-origin-when-cross-origin

You can also use the `curl` command line tool to see the headers associated with any request.

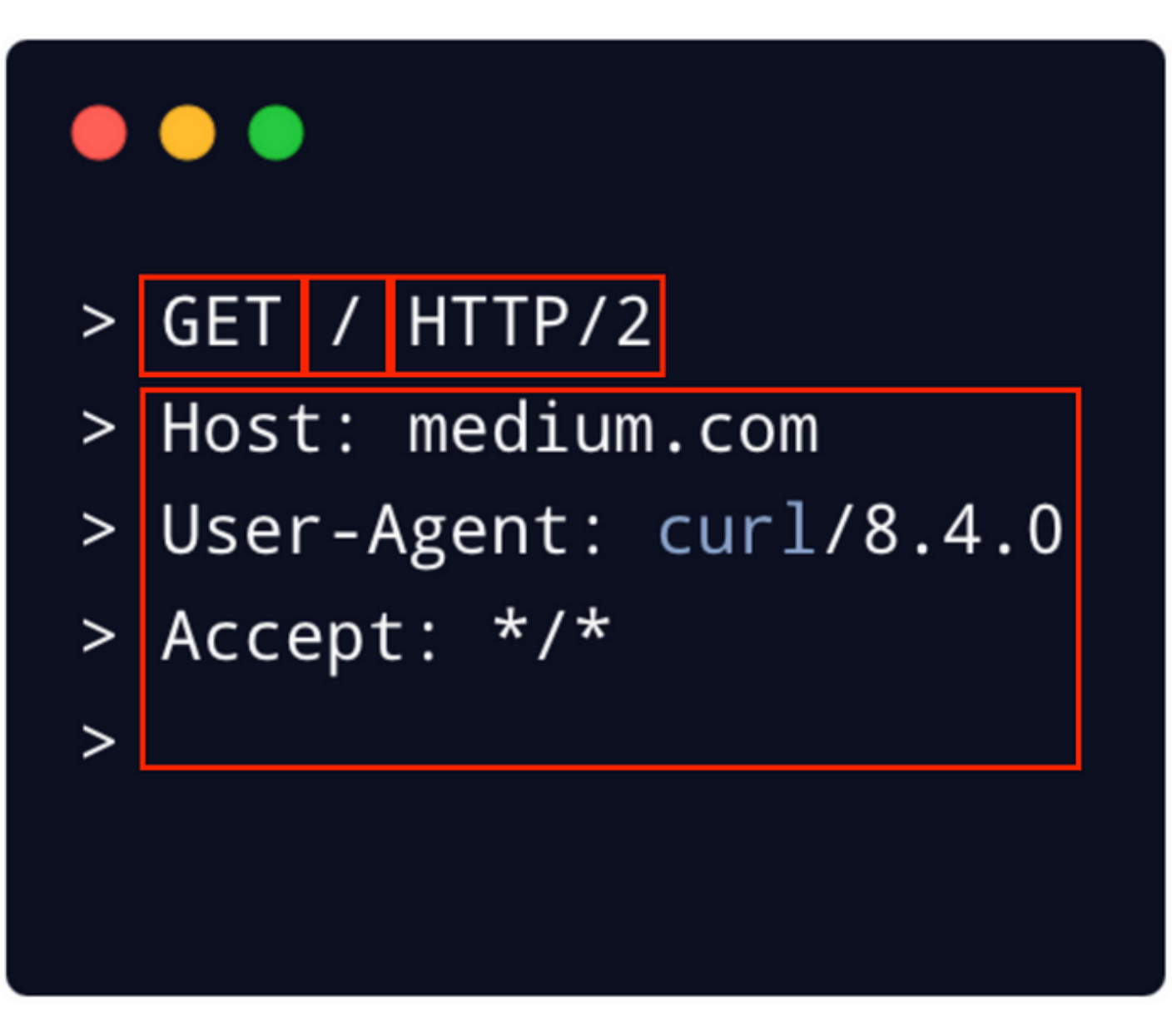
```
# View response headers only
curl -I https://example.com

# View both request and response headers
curl -v https://example.com >/dev/null
```

Then there are some third party apps like Postman that provides you with a professional UI/UX to manage your HTTP requests/response.

HTTP Request Structure

The first line of the HTTP request is called the **request line** and consists of the HTTP method, the path, and the HTTP protocol version.

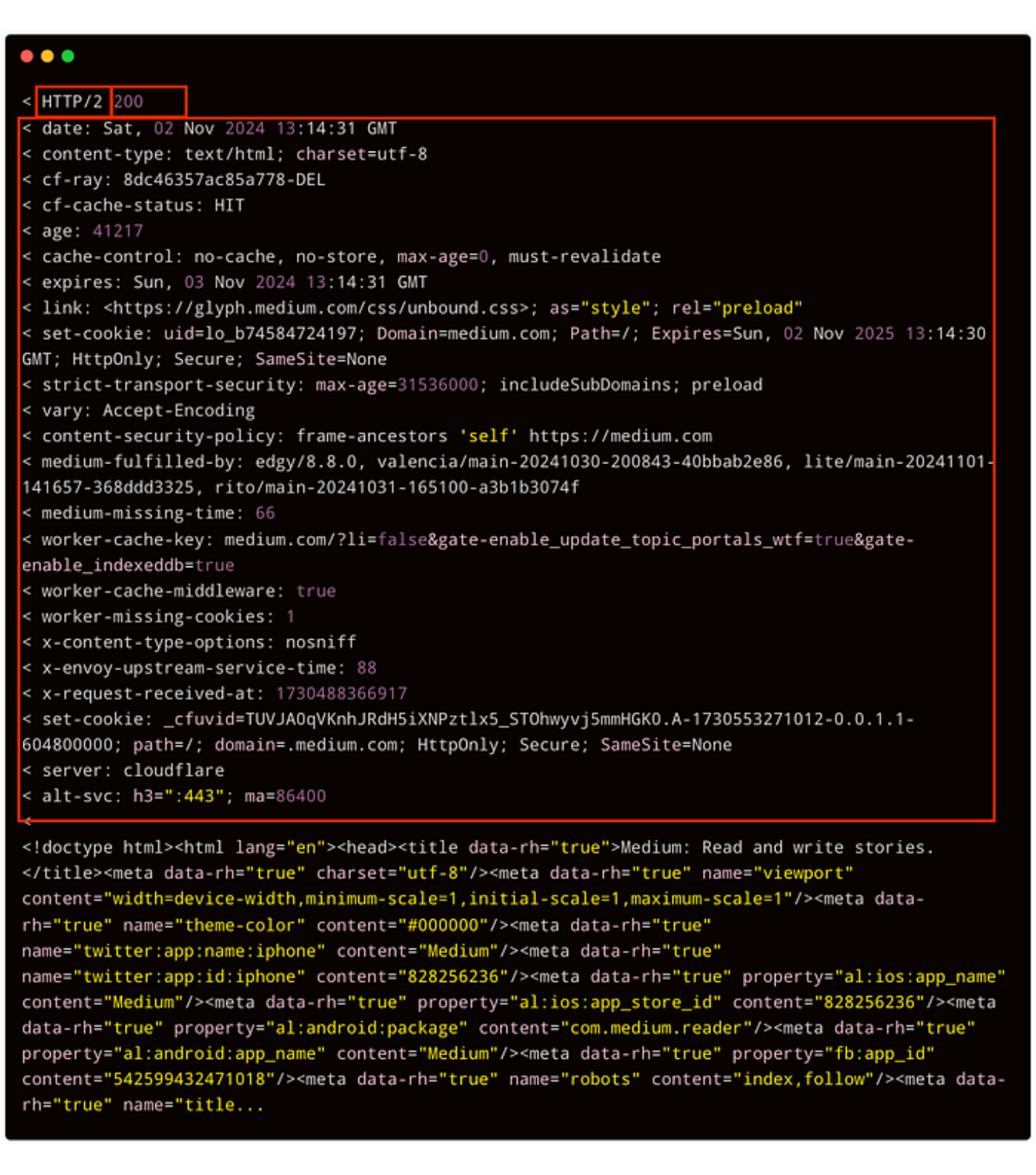
The screenshot below shows a **GET** request made to `medium.com/` using HTTP v2.0. Below that we see three request headers.



```
> GET / HTTP/2
> Host: medium.com
> User-Agent: curl/8.4.0
> Accept: */*
>
```

HTTP Response Structure

The HTTP response received from the server looks something like the following.



```
< HTTP/2 200
< date: Sat, 02 Nov 2024 13:14:31 GMT
< content-type: text/html; charset=utf-8
< cf-ray: 8dc46357ac85a778-DEL
< cf-cache-status: HIT
< age: 41217
< cache-control: no-cache, no-store, max-age=0, must-revalidate
< expires: Sun, 03 Nov 2024 13:14:31 GMT
< link: <https://glyph.medium.com/css/unbound.css>; as="style"; rel="preload"
< set-cookie: uid=lo_b74584724197; Domain=medium.com; Path=/; Expires=Sun, 02 Nov 2025 13:14:30
GMT; HttpOnly; Secure; SameSite=None
< strict-transport-security: max-age=31536000; includeSubDomains; preload
< vary: Accept-Encoding
< content-security-policy: frame-ancestors 'self' https://medium.com
< medium-fulfilled-by: edgy/8.8.0, valencia/main-20241030-200843-40bbab2e86, lite/main-20241101-
141657-368ddd3325, rito/main-20241031-165100-a3b1b3074f
< medium-missing-time: 66
< worker-cache-key: medium.com/?li=false&gate-enable_update_topic_portals_wtf=true&gate-
enable_indexeddb=true
< worker-cache-middleware: true
< worker-missing-cookies: 1
< x-content-type-options: nosniff
< x-envoy-upstream-service-time: 88
< x-request-received-at: 1730488366917
< set-cookie: _cfuvid=TUVJA0qVKnhJRdH5iXNPztlx5_STOhwyvj5mmHGK0.A-1730553271012-0.0.1.1-
604800000; path=/; domain=.medium.com; HttpOnly; Secure; SameSite=None
< server: cloudflare
< alt-svc: h3=":443"; ma=86400
<
<!doctype html><html lang="en"><head><title data-rh="true">Medium: Read and write stories.
</title><meta data-rh="true" charset="utf-8"/><meta data-rh="true" name="viewport"
content="width=device-width,minimum-scale=1,initial-scale=1,maximum-scale=1"/><meta data-
rh="true" name="theme-color" content="#000000"/><meta data-rh="true"
name="twitter:app:name:iphone" content="Medium"/><meta data-rh="true"
name="twitter:app:id:iphone" content="828256236"/><meta data-rh="true" property="al:ios:app_name"
content="Medium"/><meta data-rh="true" property="al:ios:app_store_id" content="828256236"/><meta
data-rh="true" property="al:android:package" content="com.medium.reader"/><meta data-rh="true"
property="al:android:app_name" content="Medium"/><meta data-rh="true" property="fb:app_id"
content="542599432471018"/><meta data-rh="true" name="robots" content="index,follow"/><meta data-
rh="true" name="title...
```

You will notice that the response line shows the HTTP version used and the status code to indicate the outcome of the request.

Below the response line is a bunch of headers that are followed by the actual payload of the HTTP request.

HTTP Status Codes

Different status code values convey different meanings, and most servers today adhere to the following conventions ~

- 2xx are used for successful requests
- 3xx are for redirections
- 4xx are used if there is some problem with the request
- 5xx are used if there is some problem with the server handling the request

A complete list of all the HTTP codes can be found on Wikipedia.

Headers in HTTP Requests

Let's explore some common headers you'll encounter in an HTTP request and discuss their significance.

Host

The `Host` request header specifies the host and port number of the server to which the request is being sent. If no port is included, the default port for the service requested is implied (e.g., `443` for an HTTPS URL, and `80` for an HTTP URL).

```
Host: medium.com
```

User-Agent

The `User-Agent` request header is a characteristic string that lets servers identify the application, operating system, vendor, and/or version of the requesting client.

```
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.3
```

Using the `User-Agent` header, web-servers can collect general information about the requesting client and serve them accordingly.

Accept-Language

The `Accept-Language` header conveys the default language settings for a user. If a web-server supports different language versions for a web-page, it can redirect the client based on this header.

```
Accept-Language:en-GB,en-US;q=0.9,en;q=0.8
```

It can carry multiple languages, separated by commas. The first one is the preferred language, and each other listed language can carry a `q` value, which is an estimate of the user's preference for the language on a scale of 0 to 1.

Accept-Encoding

The `Accept-Encoding` header indicates the content encoding (usually a compression algorithm) that the client can understand.

```
Accept-Encoding: gzip, deflate, br, zstd
```

Most modern browsers support gzip and will send this in the header. The web server then can send the HTML output in a compressed format. This can reduce the size by up to 80% to save bandwidth and time.

If-Modified-Since

The `If-Modified-Since` request HTTP header makes the request conditional: the server sends back the requested resource, with a `200` status, only if it has been last modified after the given date. If the resource has not been modified since, the response is a `304` without any body.

```
If-Modified-Since: Sat, 28 Sep 2024 03:21:52 GMT
```

Cookie

The `Cookie` header sends the cookies stored in your browser for that domain.

```
Cookie: PHPSESSID=298zf09hf012fh2; csrftoken=u32t4o3tb3gg43; _gat=1
```

These are name=value pairs separated by semicolons. Cookies can also contain the session id.

Referer

The `Referer` header allows a server to identify referring pages that people are visiting from or where requested resources are being used. This data can be used for analytics, logging, optimised caching, and more.

```
Referer: <url>
```

Authorization

The HTTP `Authorization` request header can be used to provide credentials that authenticate a client with a server, allowing access to protected resources.

When a web page asks for authorization, the browser opens a login window. When you enter a username and password in this window, the browser sends another HTTP request, but this time it contains this header.

```
// Bearer
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

// Basic
Authorization: Basic base64(username:password)

// API Key
Authorization: ApiKey your-api-key
```

Headers in HTTP Response

Let us now shift our attention to some common headers in an HTTP response.

Cache Control

The `Cache-Control` HTTP header controls the browser caching behaviour. When someone visits a website, their browser will save certain resources, such as images and website data, in a store called the cache. When that user revisits the same website, cache-control sets the rules which determine whether that user will have those resources loaded from their local cache, or whether the browser will have to send a request to the server for fresh resources.

```
Cache-Control: max-age=3600, public
```

Caching can also be prevented by using the `no-cache` directive.

```
Cache-Control: no-cache
```

Content Type

The `Content-Type` header is used to indicate the media type of the responded payload. For example, if the HTTP response contains an HTML file, the `Content-Type` header will have the following value.

```
Content-Type: text/html; charset=UTF-8
```

Browser uses the `Content-Type` header to decide how to interpret the payload data.

Content Disposition

The `Content-Disposition` header is important for controlling how browser handles file downloads initiated by a web-server. It provides information about the file being downloaded, influencing whether the browser displays it inline or offers it as a download.

```
Content-Disposition: attachment; filename="download.zip"
```

A value of `inline` for `Content-Disposition` will instruct the browser to display the file inside the browser instead of downloading it.

Content-Length

The HTTP `Content-Length` header indicates the size, in bytes, of the message body sent to the recipient.

```
Content-Length: 89123
```

This is especially useful for file downloads. That's how the browser can determine the progress of the download.

Location

The `Location` header is used during re-directions. When the server responds with a 301 or 302 status, the browser will consult the `Location` header to decide the re-direction.

```
Location: /index.html
```

Set-Cookie

When a web-server wants to store certain data on the browser, it sends an HTTP request with `Set-Cookie` header included. This small amount of data is called cookie and is kept in browser's storage. All the subsequent requests to the web-server will then contain these cookies in header.

```
Set-Cookie: sessionId=38afes7a8
```

If the expiration date is not specified, the cookie is deleted when the browser window is closed.

While this article highlighted key HTTP headers, remember that a rich world of header functionality awaits. Explore the official documentation and discover how specific headers can optimise your web projects.

If you enjoyed reading this article, consider clapping and responding!

#https #web-development #software-development #network-security #interview